Script-Controlled ACL – Restrict Record Access Based on Field Value

Introduction

Access Control List (ACL) is used in Service Now to control access to records and fields. It helps to ensure that only authorized users can view or modify specific information.

In this project, a script-controlled ACL is designed to restrict record access based on the value of a particular field. The system checks the field value before allowing the user to access the record.

For example, if an Incident record has a field such as Priority, Assignment Group, or State, access can be controlled according to its value.

- Objective

The main objectives of this project are:

1. To control record access using ACL.
2. To restrict unauthorized users from viewing or modifying records.
3. To use a script to check field values dynamically.
4. To improve data security in Service Now.
5. To provide access only when the specified conditions are satisfied.

In a basic access-control system, users may have access to records based mainly on their roles.

This may not be sufficient when access needs to depend on the value stored in a particular record field.

Limitations

- Less flexible record-level security.
- Difficult to control access based on changing field values.
- Users may get access to records that they do not need.
- Proposed System

The proposed system uses a Script-Controlled ACL.

The ACL script checks the value of a selected field in the record. Based on the condition, the system allows or denies access.

Example

Suppose an Incident table contains an Assignment Group field.

-If Assignment Group = IT Support → Access is allowed.

- If Assignment Group ≠ IT Support → Access is denied.

Thus, access is controlled dynamically based on the field value.

- Project Design

The project design consists of the following components:

User → Service Now Record → ACL Check → Script Condition → Allow / Deny Access

Design Flow

1. User tries to access a record.

2. Service Now checks the applicable ACL.

3. ACL executes the configured script.

4. Script reads the required field value.

5. The field value is compared with the required condition.

6. If the condition is satisfied, access is granted.

7. If the condition is not satisfied, access is denied.

- Modules

Module 1 – User Access

The user attempts to access a particular record.

Module 2 – ACL Configuration

An ACL is created for the required table and operation such as Read, Write, Create, or Delete.

Module 3 – Field Value Validation

The ACL script checks the value of the selected field.

Module 4 – Access Decision

Based on the script result, the system either allows or denies access.

Sample Script Logic

For example, access can be allowed only when the Assignment Group is the required group.

Answer = current ,assignment group == “Required Group”;

The exact field and condition can be changed according to the project requirement.

Advantages

- Provides better record-level security.
- Prevents unauthorized access.
- Allows dynamic access control.
- Easy to customize according to field values.
- Improves data protection.
- Reduces unnecessary access to sensitive record.

Conclusion

The Script-Controlled ACL – Restrict Record Access Based on Field Value project provides a flexible method for controlling access to records.

By using ACL scripts, access can be dynamically controlled based on the values stored in records. This improves security and ensures that users can access only the records they are authorized to access.